

$$Ab_2 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 3 + 3 \cdot (-2) \\ 1 \cdot 3 + (-5) \cdot (-2) \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$= \begin{bmatrix} 0 \\ 13 \end{bmatrix}$$

$$Ab_3 = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 6 + 3 \cdot 3 \\ 1 \cdot 6 + (-5) \cdot 3 \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$= \begin{bmatrix} 21 \\ -9 \end{bmatrix}$$

$$AB = \begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \end{bmatrix} \quad \square$$

Notice that since the first column of AB is Ab_1 , this column is a linear combination of the columns of A using the entries in b_1 as weights.

Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B . Important

The definition of AB shows that AB has the same number of rows as A and the same number of columns as B .

Row-Column Rule for Computing AB

If the product AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B . If $(AB)_{ij}$ denotes the (i,j) -entry in AB , and if A is an $m \times n$ matrix, then

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

By definition of matrix multiplication, the j th column of matrix AB is AB_j . $A = [a_1 \ a_2 \ \dots \ a_n]$. By defn. of matrix eqn.

$$AB_j = a_1 b_{1j} + a_2 b_{2j} + \dots + a_n b_{nj} = AB_j$$

$\therefore$ The i th entry of AB_j is: $a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj} \quad \square$